MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 12** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If PA and PB are tangents to a circle with centre O inclined to each other at $70°$, then $\angle OAB$ is equal to: [1]

(A) $35°$ (B) $45°$ (C) $55°$ (D) $70°$

Q2. In $\triangle ABC$, $AB = 12$ cm, $BC = 16$ cm, and $\angle B = 90°$. A circle is inscribed in the triangle. The inradius of the circle is: [1]

(A) 3 cm (B) 4 cm (C) 5 cm (D) 6 cm

Q3. If a chord of length 30 cm of a circle of radius 17 cm touches a concentric circle, the radius of the concentric circle is: [1]

(A) 8 cm (B) 10 cm (C) 12 cm (D) 15 cm

Q4. A quadrilateral $ABCD$ circumscribes a circle. If $AB = x + 2$, $BC = 6$, $CD = x + 4$, and $DA = 8$, the value of x is: [1]

(A) 2 (B) 3 (C) 4 (D) 5

Q5. If tangent segments PA and PB from P to circle with centre O satisfy $PA = 8$ cm and $\angle AOB = 120°$, the length of chord AB is: [1]

(A) 8 cm (B) $8\sqrt{3}$ cm (C) $4\sqrt{3}$ cm (D) 16 cm

Q6. **Assertion (A):** If tangents PA and PB are drawn to a circle with centre O such that $\angle APB = 90°$, then quadrilateral $OAPB$ is a square.
Reason (R): A quadrilateral having three right angles and adjacent sides equal (radii $OA = OB$) is a square. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. If PA and PB are tangents to a circle with centre O from an external point P such that OP equals the diameter of the circle, prove that $\triangle PAB$ is equilateral. [2]

Q8. A quadrilateral $ABCD$ circumscribes a circle. If $AB = 2x + 3$, $BC = 7$, $CD = 3x - 1$, and $DA = 9$, find the value of x and the length of side AB. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. A circle touches all four sides of a quadrilateral $ABCD$ whose sides are $AB = x, BC = x + 3, CD = 2x - 1, DA = x + 4$. Find the value of x and the area of the circle if its radius is 3 cm. [3]

Q10. Prove that opposite angles of any quadrilateral circumscribing a circle have supplementary central projections, and if the quadrilateral is a trapezium with parallel sides $AB \parallel CD$, then $AB + CD = BC + AD$. [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

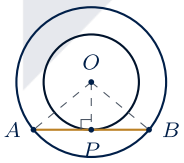
Q11. A circle with centre O is inscribed in a quadrilateral $ABCD$ touching AB, BC, CD, DA at P, Q, R, S. If $AB \perp BC$, $BC = 38$ cm, $CD = 25$ cm, $AD = 28$ cm, find: (i) the length of side AB, (ii) the radius r of the incircle, (iii) the area of quadrilateral $BPOQ$. [5]

Q12. Two tangents PA and PB are drawn to a circle with centre O from an external point P. A chord CD is drawn parallel to tangent PA such that it intersects PB at E and the circle at C and D. Prove that: (i) OP is perpendicular to AB, (ii) $\triangle PAB$ is isosceles, (iii) If $\angle APB = 40°$, find all interior angles of $\triangle PAB$ and the central angle $\angle AOB$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Circular Archery Target with Concentric Scoring Rings

An Olympic archery target consists of concentric scoring circles. The inner gold ring has radius $r = 6$ cm and the adjacent red ring has outer radius $R = 10$ cm, both centred at O. An arrow path marks a straight chord AB on the red ring that touches the boundary of the gold ring at M.



(i) State why triangle $\triangle OMA$ is a right triangle. [1]

(ii) Find the length of half-chord AM. [1]

(iii) Calculate the full length of chord AB, OR find the area of the red ring between the two concentric circles. (Use $\pi = 3.14$) [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★